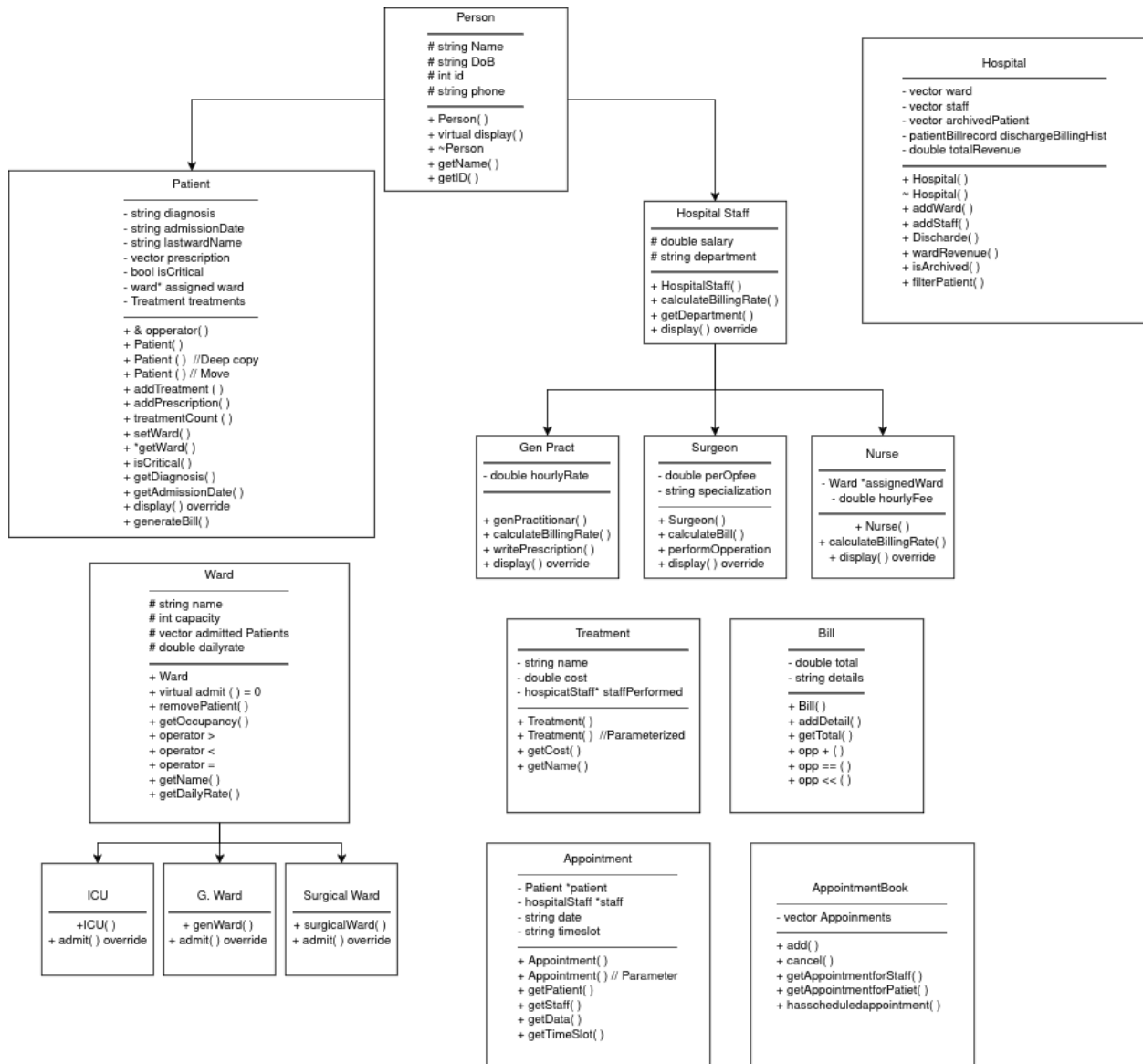


Design Document

Class Diagram and relationships:



Explanation:

1. The Operational Flow

This is how a patient moves through the code:

- **The Booking:** A patient starts in the `AppointmentBook`. The system checks (`Staff + Date + Time`) to make sure no doctor is in two places at once.
 - **The Ward:** Patients are assigned a `Ward`. We use **Logic Gates** here: if you aren't "Critical," the `ICU` won't let you in.
 - **The Care:** `Staff` (GPs, Surgeons, Nurses) are assigned. When a Surgeon performs an operation, it automatically logs a `Treatment` and its cost to that patient.
 - **The Exit:** When it's time to leave, the system calculates the **Bill** (`Stay + Treatments + Fees`).
 - **The Archive:** Once paid, the patient is **Moved** (not copied) into `archivedPatients`. This keeps the "Active" hospital list clean and the beds free.
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2. How the "Engine" Works

To make this flow work without crashing or leaking memory, we used a few specific C++ tricks:

- **The "Rule of Three":** Since we're managing memory manually, we built **Deep Copy** and **Assignment** logic for Patients. This ensures that cloning a patient doesn't accidentally link their medical records to the original.
 - **Move Semantics:** When a patient is discharged, we use `std::move`. It's like sliding a tray from one table to another instead of rebuilding the whole meal from scratch—much faster.
 - **Smart Comparisons:** We overloaded the `<` and `>` operators for Wards. This lets the hospital compare Wards based on **Occupancy %** (how full they are) rather than just their size.
 - **Custom Tools:** We avoided bulky libraries and built our own `custom_filter`. It uses **Lambdas** to quickly find things (like "all patients with heart issues") without writing a hundred different loops.
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3. Revenue Tracking

We decoupled money from the people. When a patient leaves, their financial data is snapped into a `PatientBillRecord`. This means the `Hospital` can report `wardRevenue` for "Ward A" even if every patient in that ward has already been discharged and deleted.